



COMPUTER SCIENCE HSSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Signature should not cross this line

Version No.				
3	0	0	7	1

0	●	●	0	0
1	1	1	1	●
2	2	2	2	2
●	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	●	7
8	8	8	8	8
9	9	9	9	9

ROLL NUMBER						

0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. What is the primary purpose of a scatter plot in data analysis?	To summarize categorical data	To calculate mean and median	To display distribution of a single variable	To show the relationship between two variables	☐	☐	☐	☐
2. Which computational thinking process involves breaking a complex problem into smaller, manageable parts?	Abstraction	Decomposition	Pattern Recognition	Algorithm Design	☐	☐	☐	☐
3. A company installs a system that blocks suspicious incoming traffic and prevents unauthorized users from accessing internal resources. Which security method is being used?	Encryption	Backup	Firewall	Compression	☐	☐	☐	☐
4. In black box testing, the tester mainly focuses on:	Internal logic of the program	Source code structure	Functional requirements and expected outputs	Memory management	☐	☐	☐	☐
5. Why should one keep their personal information private online?	To make the profile look mysterious	To protect from identity theft or scams	To make it harder for friends to contact	To reduce storage on device	☐	☐	☐	☐
6. Which of the following is a reserved keyword in Python?	while	input	print	sum	☐	☐	☐	☐
7. When refining a computational artifact, what is the primary goal?	To rewrite the pseudocode in another language	To remove all comments from the program	To avoid using loops and functions	To improve efficiency and correctness of solution	☐	☐	☐	☐
8. A researcher talks to students one-on-one to understand their feelings about online classes. What method is being used?	Survey	Interview	Simulation	Observation	☐	☐	☐	☐
9. Which stakeholder value is most directly related to ensuring fairness in AI systems?	Profit maximization	Hardware efficiency	Cultural diversity	Faster internet speed	☐	☐	☐	☐
10. Why is building a prototype considered a critical step in business idea development?	It transforms abstract ideas into a workable model	It eliminates the need for future improvements	It guarantees market success instantly	It avoids the need for design skills	☐	☐	☐	☐

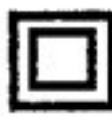


	Question	A	B	C	D	A	B	C	
11.	What is a key cultural and human impact of increased connectivity?	No change in cultural practices	Complete cultural isolation	Elimination of human interaction	Better cultural exchange	☐	☐	☐	☐
12.	What will be the output of the following statement? <pre>for i in range(3): print(i)</pre>	0 1 2	1 2 3	0 1 2 3	3 2 1	☐	☐	☐	☐
13.	A group of students builds a physical model of an eco-friendly water filter. During testing, they notice it does not filter water efficiently. What is the next step in the prototyping process?	Present the model to the class without changes	Ignore the problem and continue with other projects	Throw away the prototype completely	Adjust the design, test again, and improve the filter	☐	☐	☐	☐
14.	Which principle ensures that experimental results are not influenced by biases and that each subject has an equal chance of receiving different treatments?	Randomization	Replication	Blocking	Observation	☐	☐	☐	☐
15.	A company wants to create a mobile app for online food ordering. The development team prepares detailed diagrams showing the screens, user interactions, and data flow before coding begins. Which activity is the team performing?	Coding	Design	Testing	Deployment	☐	☐	☐	☐
16.	Which source is considered reliable for academic research?	A peer-reviewed journal article	A random blog with no author credentials	A social media post without references	A forum comment	☐	☐	☐	☐
17.	A company wants to predict future sales based on past advertising spending. Which type of model is most appropriate?	Classification model	Regression model	Clustering model	Decision tree model	☐	☐	☐	☐

—1HA-I 26007-(30071) (B) —

FBISE Past Papers

ROLL NUMBER					





COMPUTER SCIENCE HSSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

52

Time allowed: 2:35 Hours

Total Marks Sections B and C: 63

SECTION – B (Marks 39)

Q. 2 Answer the following parts briefly.

(13 x 3 = 39)

(i)	Identify and correct any three errors from the following Python code segment. <pre>num1 = input("Enter first number") num2 = input("Enter second number") sum = num1 + num2 print("Sum = " sum</pre>	1x3	OR	Construct a trace table that records the values after each iteration of the loop, and then writes the final output produced by the pseudocode. Step 1: Initialize i = 2 Step 2: While i <= 10 Step 3: Print i Step 4: Increment i by 2 Step 5: End While	03
(ii)	State the purpose of a use case when applying statistical models to real-world problems. Give an example.	2+1	OR	What is cloud computing? Explain public cloud with example.	1+2
(iii)	Describe the importance of conducting a feasibility study. How it influences decision-making in system development?	1.5 + 1.5	OR	Differentiate between training a model with supervised learning and unsupervised learning, using a daily life example.	1.5 + 1.5
(iv)	Why is prototype testing important in the product development process? Give three reasons.	1x3	OR	How do bus and star topologies differ in terms of cost, fault tolerance, and architecture?	1x3
(v)	Assume the variables x, y and z where x = 4, y = 8, and z = 2. What value will be stored in result1, result2 and result3 after executing following Python statements? <pre>result1 = x + y result2 = z ** 4 result3 = y / x</pre>	1x3	OR	For each of the following scenarios, decide whether the Waterfall model or the Agile model would be more suitable for software development. Justify with reason: a. Building construction b. Developing web application c. Digital marketing campaign	03
(vi)	How the digital divide affects human interaction and cultural participation in a connected world? Give reason.	1.5 + 1.5	OR	What is meant by a stakeholder in the context of AI system design? Why is it important to consider their culture and values?	1+2
(vii)	Which search algorithm Linear or Binary Search is more appropriate for finding a student's roll number in a sorted list? Justify your answer.	1+2	OR	Two students solve the same problem in Python: one uses global variables inside function. While the other passes values as parameters and returns the results. Which approach is better, and why?	1+2
(viii)	Evaluate the following expression using the correct order of operations in Python: $10 / 2 * 3 + 4 \% 3 - 2 ** 2$	03	OR	Draw OR gate with its truth table.	1+2
(ix)	Write down the role of simulation in research and give one example.	2+1	OR	Explain what is meant by a product and a service, with one example each.	1.5 + 1.5
(x)	Construct the logic circuit diagram for the given Boolean expression. $F = ABC + \bar{A}\bar{B}C + \bar{A}B\bar{C}$	03	OR	Write a Python program that takes an alphabet as input and prints whether it is a lower case or upper case.	03
(xi)	What is the importance of qualitative interviews in collecting research data? Explain briefly.	03	OR	Explain briefly two harmful environmental effects of increased connectivity, providing suitable examples.	1.5+ 1.5
(xii)	How does vertical scalability differ from horizontal scalability in cloud computing? Illustrate with three differences.	1x3	OR	Write a Python function named welcome() that takes a user's name "Ali" as a parameter and prints a welcome message.	03
(xiii)	Write a Python program using a for loop to print numbers from 1 to 10.	03	OR	What is the difference between parameter and statistic? Illustrate with an example.	2+1

SECTION – C (Marks 24)

Attempt the following questions.

(4 x 6 = 24)

Q.3	Simplify the Boolean Function F, using Karnaugh Map. $F = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + ABC$ Also construct a logic circuit for simplified expression.	3+3	OR	Write a Python program to find the smallest of three numbers using function.	06						
Q.4	Differentiate between primary and secondary data collection methods with example. Discuss their role in research with respect to authenticity, cost, and reliability.	3+3	OR	Why is cybersecurity essential in today's digital world? How different cybersecurity measures help protect sensitive information from unauthorized access? Provide any four in measures.	2+4						
Q.5	Sort the following numbers in ascending order using Bubble sort algorithm: <table border="1"><tr><td>17</td><td>89</td><td>26</td><td>42</td><td>3</td><td>38</td></tr></table>	17	89	26	42	3	38	06	OR	Develop a Python program that calculates the product of even numbers from 1 to 20 using a loop.	06
17	89	26	42	3	38						
Q.6	Explain the importance of blockchain technology. Also describe two technologies that form its foundation.	2+4	OR	What is meant by the safe use of information sources? Explain any five safe practices that ensure information is used responsibly.	1+5						



COMPUTER SCIENCE HSSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Signature should not cross this line

Version No.				
3	2	0	7	5

0	0	●	0	0
1	1	1	1	1
2	●	2	2	2
●	3	3	3	3
4	4	4	4	4
5	5	5	5	●
6	6	6	6	6
7	7	7	●	7
8	8	8	8	8
9	9	9	9	9

ROLL NUMBER						

0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. A company wants to develop a library management system. They interview librarians to understand the current process, identify problems, and gather requirements for the new system. Which activity are the developers performing?	Writing code for the system	Testing the completed system	Analyzing user needs and system requirements	Designing the system interface	☐	☐	☐	☐
2. What is a safe practice when using information from the internet?	Using information without checking its accuracy	Relying only on personal opinions	Sharing information everywhere	Checking the website's credibility	☐	☐	☐	☐
3. A student is analyzing the relationship between hours studied and exam scores. What does a regression model help them understand?	Which students are failing	How to store data efficiently	The numerical relationship to make predictions	How to group students by scores	☐	☐	☐	☐
4. Why is a bar chart commonly used in data analysis?	To show the trend of data over time	To display the spread of continuous data	To calculate correlation between variables	To compare values across different categories	☐	☐	☐	☐
5. Which computational thinking process focuses on identifying similarities or trends among problems?	Decomposition	Pattern Recognition	Simulation	Abstraction	☐	☐	☐	☐
6. An organization converts sensitive data into coded form so that even if hackers access it, they cannot understand it. Which method is being used?	Encryption	Antivirus	Debugging	Firewall	☐	☐	☐	☐
7. Which of the following tasks best represents white box testing?	Checking how the program behaves for different inputs	Analyzing internal code structure and logic paths	Verifying output without knowing implementation	Testing the system using only user requirements	☐	☐	☐	☐
8. Which behavior shows good digital etiquette?	Respecting others' opinions and replying politely	Ignoring rules of online platforms	Downloading illegal content	Sending messages in all capital letters to show importance	☐	☐	☐	☐



	Question	A	B	C	D	A	B	C	D
9.	In Python, which data type is used to represent text?	float	str	bool	int	☐	☐	☐	☐
10.	A student writes pseudocode for a program but finds errors after running the code. What should the student do next?	Refine the pseudocode and retest the program	Rewrite the program in another language without testing	Remove all comments from the program	Skip testing and submit the program	☐	☐	☐	☐
11.	Why are qualitative interviews considered flexible?	They only collect yes/no answers	They allow participants to share ideas in their own words	They require no preparation	They always follow a strict questionnaire	☐	☐	☐	☐
12.	Which blockchain type allows anyone to join and participate?	Hybrid blockchain	Consortium blockchain	Public blockchain	Private blockchain	☐	☐	☐	☐
13.	Which of the following best describes an MVP?	A prototype with no real users	A basic product with no functionality	A basic version with essential functionality	A product with all planned features	☐	☐	☐	☐
14.	How does increased connectivity affect global cultures?	Keeps cultures separate	Prevents communication between countries	Removes all cultural differences	Promotes cultural interaction	☐	☐	☐	☐
15.	What is the output of the following code? <code>for i in range(3): print(i*2)</code>	0 2 4	2 4 6	1 2 3	0 1 2	☐	☐	☐	☐
16.	Sara has designed a mobile app prototype that helps people find local farmers' markets. After letting her friends use it, she notices some buttons are confusing. What should Sara do next?	Delete the prototype	Only write a report about the app	Refine the app and retest	Launch the app immediately	☐	☐	☐	☐
17.	Why is replication important in experimental design?	To make the experiment more complex	To reduce the number of variables in the experiment	To eliminate the need for a control group	To ensure that results are consistent and reliable across multiple trials	☐	☐	☐	☐

—1HA-I 26007-32075 (D)—

ROLL NUMBER						





COMPUTER SCIENCE HSSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

564

Time allowed: 2:35 Hours

Total Marks Sections B and C: 63

SECTION – B (Marks 39)

2. Answer the following parts briefly.

(13 x 3 = 39)

	Identify and correct any three errors from the following Python code segment. <code>length = input("Enter length") width = input(Enter width") area = length * width print("Area of rectangle is:" area</code>	03	OR	Construct a trace table that records the values after each iteration of the loop, and then write the final output produced by the pseudocode. Step 1: For i from 1 to 5 Step 2: Calculate square = i * i Step 3: Print square Step 4: End For Step 5: Stop	03
(i)	List the main steps of predictive analysis that support the use case of statistical modeling in weather forecasting.	03	OR	Briefly explain private and public cloud deployment models with examples.	1.5 + 1.5
(ii)	A university is introducing a new student information system. Using this case study, recommend the most suitable implementation method of SDLC. Justify your recommendation by outlining two considerations.	1+2	OR	Differentiate between supervised and unsupervised learning with the help of daily life example.	1.5 + 1.5
(v)	Why is prototype performance testing important before moving to full-scale production? Support your answer with three reasons.	1x3	OR	Compare star and ring topologies with respect to scalability, reliability, and architecture.	1x3
(v)	A Python program is written to calculate the total and average marks. In one version, all the logic is written in a single block without using functions, while in another version, the program is divided into smaller functions. Which approach is better, and why?	1+2	OR	For each of the following scenarios, decide whether the Waterfall model or the Agile model would be more suitable for software development. Justify with reason. a. Automobile production b. Public works projects c. Research projects	1x3
(vi)	How increased connectivity through computing has influenced cultural exchange and human interaction globally? Explain briefly.	1.5 + 1.5	OR	Write down three positive impacts of AI systems.	1x3
(vii)	Which search algorithm Linear or Binary Search is more appropriate for finding a Book ID in an unsorted library catalog? Justify your answer.	1+2	OR	Assume the variables a = 6, b = 4, and c = 2. What value will be stored in x, y, and z after executing the following Python statements? a. x = a + b b. y = x - c c. z = y * b	1x3
(viii)	Evaluate the following expression in Python using the proper order of operations: <code>8 % 5 + 6 // 2 * 3 - 2 ** 3</code>	03	OR	Draw NAND gate with its truth table.	1+2
(ix)	What is the importance of using visuals (like charts or graphs) in a report? Explain any three points.	1x3	OR	Write a Python function named <code>add_numbers()</code> that takes two numbers 4 and 6 as parameters and prints their sum.	03
(x)	Draw the logic circuit that corresponds to the following Boolean expression. $F = A\bar{B} + AC + \bar{B}C$	03	OR	Write a Python program that takes three numbers as input and prints which one is largest.	03
(xi)	List any three common uses of simulation that improve daily life activities.	1x3	OR	What is meant by assistive technology? Give two examples used by people with disabilities.	1+2
(xii)	If a company's network frequently crashes when more users connect, what does this indicate about its reliability and scalability? Briefly explain.	1.5 + 1.5	OR	What is the difference between correlation and causation? Illustrate with example.	1.5 + 1.5
(xiii)	Write a Python program using a while loop to print numbers from 1 to 10.	03	OR	What is the importance of successful product development? Explain any three points.	1x3

SECTION – C (Marks 24)

Attempt the following questions.

(4 x 6 = 24)

Q.3	Simplify the Boolean Function F, using Karnaugh Map. $F = A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C$ Also construct a logic circuit for simplified expression.	3+3	OR	Write a Python program with a function to find the square and cube of a number.	06						
Q.4	Explain the importance of primary data collection methods. Compare interviews and observations with respect to strengths, weaknesses, time requirements, and accuracy of results.	2+4	OR	Describe encryption as a security technique used in digital communication. Compare symmetric and asymmetric encryption methods discussing their strengths, weaknesses, key length, and large data handling in secure communication.	2+4						
Q.5	Sort the following numbers in ascending order using Insertion sort algorithm: <table border="1" style="display: inline-table; vertical-align: middle;"><tr><td>23</td><td>4</td><td>11</td><td>8</td><td>37</td><td>15</td></tr></table>	23	4	11	8	37	15	06	OR	Write a Python program that calculates the sum of following series using repetition: $1+3+5+\dots+99$	06
23	4	11	8	37	15						
Q.6	What is the importance of Internet of Things? Explain any two technologies that enabled IoT.	2+4	OR	What is meant by reliable information? Explain four sources that can be considered reliable for academic or professional uses.	2+4						

— 1HA-I 26007 (D) —